

Concept and realization of Kitaev quantum spin liquids

Hidegori Takagi^{1,2,3,*}, Tomohiro Takayama^{1,2,*}, George Jackeli^{1,2,*}, Giniyat Khaliullin^{1,*} and Stephen E. Nagler^{4,*}

Abstract | The Kitaev model is an exactly solvable $S = 1/2$ spin model on a 2D honeycomb lattice, in which the spins fractionalize into Majorana fermions and form a topological quantum spin liquid (QSL) in the ground state. Several complex iridium oxides, as well as α - RuCl_3 , are magnetic insulators with a honeycomb structure, and it was noticed that they accommodate essential ingredients of the Kitaev model owing to the interplay of electron correlation and spin–orbit coupling. This has led to a race to realize the Kitaev QSL and detect signatures of Majorana fermions. We summarize the theoretical background of the Kitaev QSL ground state and its realization using spin–orbital entangled $J_{\text{eff}} = 1/2$ moments. We provide an overview of candidate materials and their electronic and magnetic properties, including Na_2IrO_3 , α - Li_2IrO_3 , β - Li_2IrO_3 , γ - Li_2IrO_3 , α - RuCl_3 and $\text{H}_3\text{LiIr}_2\text{O}_6$. Finally, we discuss experiments showing that $\text{H}_3\text{LiIr}_2\text{O}_6$ and α - RuCl_3 in an applied magnetic field exhibit signatures of the QSL state and that α - RuCl_3 has unusual magnetic excitations and thermal transport properties consistent with spin fractionalization.

In conventional magnetic materials, interactions between the spins lead to a phase transition from a high-temperature disordered state to a magnetically ordered state as the temperature is lowered. The transition is typically accompanied by singularities in the thermodynamic observables at the transition point and spontaneous symmetry breaking and a reduction in the spin entropy to zero as the system enters a unique ground state. However, the spin entropy can also be released without any symmetry breaking, down to zero temperature, by forming a collective quantum spin state with long-range quantum entanglement. This exotic state of matter is called a quantum spin liquid (QSL)^{1–3}. A goal of condensed matter physics is to discover new quantum phases formed by the ensemble of interacting spins and charges in solids. The QSL is perhaps one of the most exotic quantum phases known so far, partly because of its nontrivial elementary excitations, and has been attracting the attention of condensed matter scientists for several decades.

The exploration of QSLs in more than one dimension was launched by the conjecture of the resonant valence bond (RVB) state by Anderson in 1973 (REF.⁴). When antiferromagnetically interacting Heisenberg spins with spin $S = 1/2$ are placed on a triangular lattice, the interactions on different bonds are frustrated owing to the geometry; intuitively, this should prevent spins from finding a unique way to break symmetry. Anderson

proposed that the ground state of the system consists of a quantum superposition of spin singlets formed by pairs of $S = 1/2$ spins, in which the paired spins fluctuate in a liquid-like fashion, unlike in magnetically ordered ground states, in which the spins are static, conceptually resembling a solid. This model gives an intuitive image of the QSL state. The ground state of the $S = 1/2$ Heisenberg antiferromagnet on the triangular lattice was later shown to be ordered: the non-collinear 120° structure^{5,6}. Nevertheless, RVB-QSLs are believed to exist in other geometrically frustrated lattices such as the kagome lattice⁷ or the triangular lattice with additional interactions^{8–10}.

A hallmark of RVB-QSLs is the emergence of unusual elementary excitations described by mobile fermionic $S = 1/2$ quasiparticles, called spinons¹, as in the 1D $S = 1/2$ Heisenberg antiferromagnet¹¹. Excitations in conventional magnets have $S = 1$ and generally show up in scattering experiments as peaks that are sharp in energy for a given momentum, that is, as magnons with a well-defined dispersion. In RVB-QSLs, the conventional $S = 1$ excitation ‘fractionalizes’ into spinon pairs. Spinons give rise to an energy continuum of excitations at a given momentum, much like electronic excitations in metals. If the spinon excitations are gapless, a Fermi surface of spinons may emerge^{12,13}.

A number of $S = 1/2$ triangular and kagome Heisenberg antiferromagnets have been shown experimentally

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Key points

- The quantum spin liquid is an exotic state of matter in which interacting spins avoid symmetry-breaking phase transitions and form a ground state exhibiting long-range entanglement and topological order.
- In the exactly soluble Kitaev model, anisotropic spin-pair interactions on different honeycomb lattice bonds conflict, producing strong frustration and a spin-liquid ground state.
- In some transition-metal-based Mott insulators, unquenched orbital moments and spin-orbit entangled wavefunctions can result in a low-energy Hamiltonian with bond-dependent interactions similar to the Kitaev model.
- Quantum spin liquid behaviour was recently discovered in the hydrogenated iridate $\text{H}_3\text{LiIr}_2\text{O}_6$, and signatures of the spin fractionalization and Majorana fermions predicted by the Kitaev model have been detected in $\alpha\text{-RuCl}_3$.

to be materializations of RVB-QSL states, including organic charge-transfer salts, κ -(bis(ethylenedithio)tetrathiafulvalene) $_2\text{Cu}_2(\text{CN})_3$ (REF.¹⁴), $\text{EtMe}_3\text{Sb}[\text{Pd}(1,3\text{-dithiole-2-thione-4,5-dithiolate})_2]_2$ (REF.¹⁵) and herbertsmithite $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ (REF.¹⁶). These lack a clear signature of magnetic ordering down to the lowest temperature measured, which is at least two orders of magnitude smaller than the energy scale of magnetic interactions. In the organic salts, a finite density of zero-energy excitations is experimentally observed as a specific heat that is linear in temperature T at low temperatures¹⁷. The associated excitations are highly mobile, as shown by a thermal conductivity linear in T (REF.¹⁸). In herbertsmithite, the presence of continuum excitations with a small excitation gap was recently demonstrated using inelastic neutron scattering (INS)¹⁹ and NMR²⁰ measurements. These are very likely the fractionalized excitations expected for a QSL.

Despite impressive progress, research on the RVB-QSL state was constrained by the difficulty that it had never been obtained as an exact solution of any realistic model Hamiltonian. However, in 2006, a theoretical breakthrough in the field of QSLs was reported. Kitaev proposed a simple new model that is exactly solvable and that gives a QSL ground state, in which the spins fractionalize into emergent quasiparticles — Majorana fermions²¹. Soon after, a spin-orbital $J_{\text{eff}} = 1/2$ Mott insulator was identified in a complex iridium oxide²². This discovery led to a theoretical proposal²³ for the realization of the Kitaev model using $J_{\text{eff}} = 1/2$ pseudo-spins in an iridate and initiated a search for the QSL state and the hidden Majorana fermions in a family of iridium and ruthenium compounds. A new interdisciplinary field emerged, comprising quantum magnetism, topological physics, correlated electron physics and solid-state chemistry.

In this Review, we summarize the theoretical background of the Kitaev QSL ground state and its realization using spin-orbital entangled $J_{\text{eff}} = 1/2$ moments. We provide an overview of candidate materials and their electronic and magnetic properties, including Na_2IrO_3 , $\alpha\text{-Li}_2\text{IrO}_3$, $\beta\text{-Li}_2\text{IrO}_3$, $\gamma\text{-Li}_2\text{IrO}_3$, $\alpha\text{-RuCl}_3$ and $\text{H}_3\text{LiIr}_2\text{O}_6$. Finally, we discuss experiments showing that $\text{H}_3\text{LiIr}_2\text{O}_6$ and $\alpha\text{-RuCl}_3$ in an applied magnetic field exhibit signatures of the QSL state and that $\alpha\text{-RuCl}_3$ has unusual magnetic excitations and thermal transport properties consistent with spin fractionalization.

The Kitaev model and quantum spin liquid

The Kitaev model²¹ consists of $S = 1/2$ spins on a honeycomb lattice, with nearest-neighbour Ising interactions with bond-dependent easy axes parallel to the x , y or z axes²¹ (FIG. 1a). The Hamiltonian of the Kitaev model is

$$H = - \sum_{\langle ij \rangle_\gamma} K_\gamma S_i^\gamma S_j^\gamma \quad (1)$$

where $\langle ij \rangle_\gamma$ is a $\gamma = x$ -type, y -type or z -type bond and the summation is taken over all honeycomb bonds. The bond-indexed coupling constants K_γ may differ on different types of bonds. The orthogonal anisotropies of the three nearest-neighbour bonds of each spin conflict with each other, leading to frustration.

In the classical limit, in which quantum mechanical $S = 1/2$ spins are replaced by a vector, the ground state manifold of the Kitaev model is extensively degenerate^{24,25}. This extensive degeneracy is a hallmark of frustration and is inherent to geometrically frustrated magnets such as kagome²⁶ and pyrochlore²⁷ Heisenberg antiferromagnets, the candidates for an RVB-QSL. In each configuration within the classical ground-state manifold of the Kitaev model, the honeycomb lattice is decomposed into non-overlapping ‘happy’ and ‘unhappy’ sets of nearest-neighbour bonds. The happy bonds are those for which spins are aligned along the corresponding easy axis. The set of happy bonds maximizes the exchange energy gain. The remaining unhappy bonds are frustrated and do not contribute to the energy of the system. The ground-state degeneracy is related to the number of ways to distribute the happy bonds on a honeycomb lattice and the two possible spin alignments on each happy bond.

When quantum mechanical effects are turned on, the system starts tunnelling and floating within the classical ground state manifold, forming a highly entangled QSL state supporting fractionalized excitations. The QSL ground state can be characterized as a quantum-mechanical superposition of the classical configurations, each having $1/3$ happy and $2/3$ unhappy bonds (FIG. 1b), reminiscent of an RVB state. The Kitaev model is a pathway to a QSL that is distinct from others because it is based on Ising, rather than Heisenberg, interactions and is exactly solvable for all K_γ (REF.²¹). The model is characterized by infinitely many local conserved quantities, the integrals of motion. The conserved quantities are flux operators defined around each hexagon as a product of six spin operators S_i^γ , with γ matching the index of the out-going bond, for instance, $W_{1-6} = 2^6 S_1^z S_2^x S_3^y S_4^z S_5^x S_6^y$ (FIG. 1a). The flux operators have quantized eigenvalues $W_{l-m} = \pm 1$ and commute with the Hamiltonian and each other. Thus, each many-body eigenstate can be labelled by the conserved flux through each hexagon, thereby block-diagonalizing the Hamiltonian.

The exact solution of the Kitaev model has a QSL ground state and is found by fractionalizing the spin degrees of freedom by expressing $S = 1/2$ operators in terms of four flavours of Majorana fermions²¹ (FIG. 1c,d). Majorana fermions are neutral self-adjoint fermions that are simultaneously particle and antiparticle.

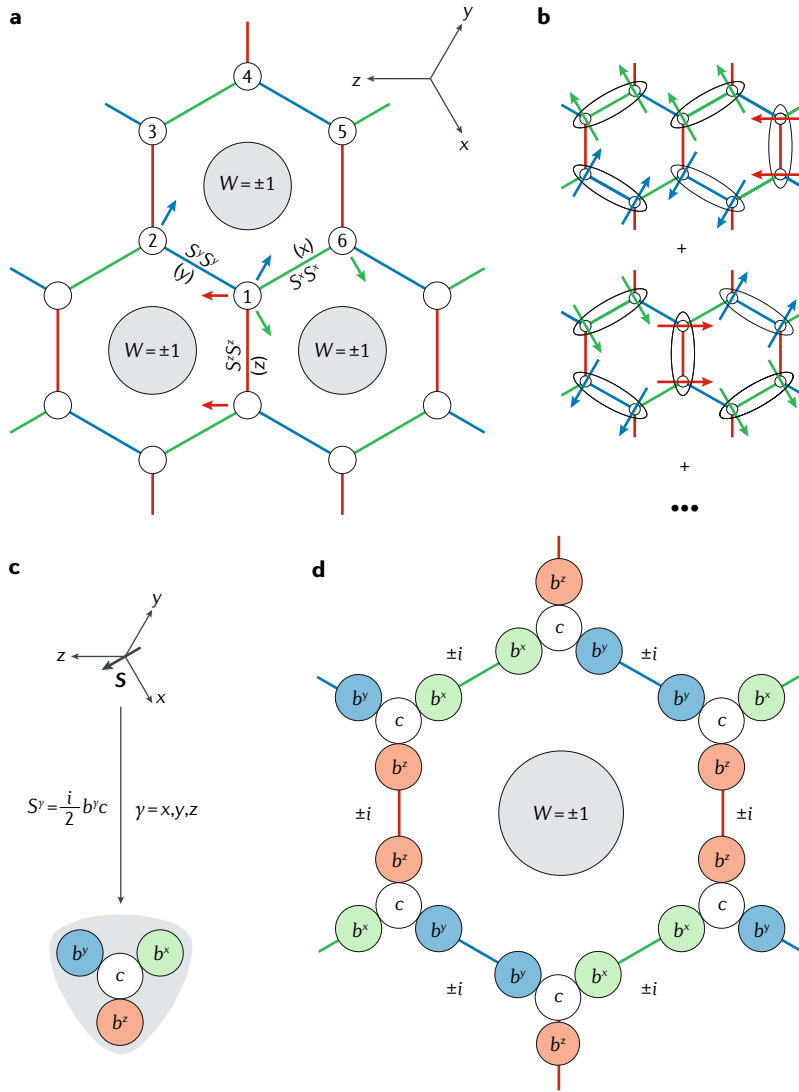


Fig. 1 | The Kitaev model. **a** | The $S = 1/2$ spins on a honeycomb lattice with bond-dependent Ising interactions. The green bonds have easy axes parallel to the x axis, the blue to the y axis and the red to the z axis. The product of six spins around each hexagonal loop forms a conserved quantity $W_{l-6} = 2^6 S_1^x S_2^z S_3^x S_4^z S_5^x S_6^z$ with eigenvalues $W = \pm 1$, defining Z_2 flux through the hexagons. **b** | A Kitaev quantum-spin-liquid state can be thought of as a quantum superposition of entangled states with different distributions of spin-paired 'happy' bonds. **c** | A real spin fractionalizes into two kinds of Majorana fermions. b^γ Represents the three localized Majorana fermions, and c represents the one itinerant Majorana fermion. **d** | Emergent Majorana fermions on the honeycomb lattice. The solid lines depict the conserved bond variables $u_{ij}^\gamma = b_j^\gamma b_i^\gamma$ of localized fermions with their imaginary eigenvalues $\pm i$, which control the phase of the c fermion hopping amplitude. Their product around each hexagonal loop measures the Z_2 flux as $W_{l-6} = u_{12}^x u_{23}^y u_{34}^z u_{45}^x u_{56}^y u_{67}^z$, $\gamma = x, y, z$.

They can be constructed from the real or imaginary part of more common complex fermions. Hence, one complex fermion mode, described by operators a and $a^\dagger$, gives rise to two Majorana modes, for example, $f_1 = (a + a^\dagger)$ and $f_2 = i(a - a^\dagger)$. Spin fractionalization — making the substitution $S_j^\gamma = \frac{i}{2} b_j^\gamma c_j$, where b_j^γ and c_j represent four Majorana modes with the constraint $b_j^x b_j^y b_j^z c_j = 1$ — preserves both the $S = 1/2$ algebra and the local 2D Hilbert space. Most importantly, this choice of Majorana representation transforms the Kitaev model into a fermionic form that explicitly and conveniently

reflects the flux-operator conservation law, key for the exact solution. In other words,

$$H = -\frac{1}{4} \sum_{\langle ij \rangle (\gamma)} K_\gamma u_{ij}^\gamma c_i c_j \quad (2)$$

where the bond operators $u_{ij}^\gamma = b_j^\gamma b_i^\gamma$ have eigenvalues $\pm i$ and their product around a hexagon determines the flux $W_{l-6} = \pm 1$. The operators u_{ij}^γ commute with the Hamiltonian and therefore are conserved. The Majorana fermions associated with b^γ are immobile owing to this conservation but control the sign of hopping of the Majorana fermions associated with c (FIG. 1d). u_{ij}^γ is an emergent Z_2 gauge field and determines the phase of the nearest-neighbour tunnelling integral of c -Majoranas, often termed matter fermions (FIG. 1d). In each flux sector, the gauge is fixed, and the operators u_{ij}^γ can be replaced by numbers $\pm i$.

The ground state is flux-free, that is, all u_{ij}^γ are equal, and matter fermions can coherently propagate through the honeycomb lattice, minimizing their kinetic energy. The corresponding dispersion is obtained by diagonalizing the quadratic (non-interacting) fermionic Hamiltonian after setting all $u_{ij}^\gamma = i$. The spectrum is of Dirac type²⁸, with all states appearing in pairs corresponding to positive and negative eigenvalues: $E_k = \pm \sqrt{\epsilon_k^2 + \Delta_k^2}$, where $\epsilon_k = 2[K_z - K_x \cos(\mathbf{k} \cdot \mathbf{a}) - K_y \cos(\mathbf{k} \cdot \mathbf{b})]$, $\Delta_k = 2[K_x \sin(\mathbf{k} \cdot \mathbf{a}) + K_y \sin(\mathbf{k} \cdot \mathbf{b})]$, where $\mathbf{k}$ denotes quasi-momentum vectors and $\mathbf{a}$ and $\mathbf{b}$ are honeycomb lattice vectors. The spectrum is gapless for weakly anisotropic coupling constants; a gap opens when one of the couplings becomes larger than the sum of the remaining two. In the gapless phase, the Dirac point can acquire a finite gap by time-reversal symmetry-breaking perturbations. For example, an external magnetic H field induces a Majorana gap $\Delta_M \approx H_x H_y H_z / K^2$, in which the exchange couplings are set equal ($K_x = K_y = K_z = K$)²¹.

The elementary excitations predicted for the Kitaev QSL mirror the fractionalization of spins, as discussed in this Review. Following the original exact solution, various physical observables such as the dynamic spin structure factor²⁹ and Raman response^{30,31} have been computed exactly. The low-energy spin excitations are localized because they consist of not only itinerant but also immobile Majorana fermions, which emerge as a Q -independent spin response with an excitation gap in the spin structure factor²⁹. The low-energy Raman response captures the fermionic excitations as the result of fractionalization. Thanks to a newly implemented numerical algorithm, especially designed for Majorana systems, the thermodynamics of the model have become accessible over almost the full temperature range³². In the specific heat $C(T)$, the fractionalization of spins appears as the presence of two well-separated peaks: one at a high temperature corresponding to the itinerant Majorana fermions and the other at a low temperature corresponding to flux ordering of the localized Majorana fermions. Each peak carries an entropy of $50\% \ln 2$. A half-quantized thermal Hall conductivity κ_{xy} , associated with the chiral edge state of Majorana fermions, $\kappa_{xy}/T = \pi/6 (k_B^2/\hbar)$, has been theoretically predicted^{21,33},

where z is a half-integer number and k_B and h are the Boltzmann and Planck constants, respectively.

Realization of the Kitaev model

Spin-orbital entangled $J_{\text{eff}} = 1/2$ moment. The Kitaev model was originally considered a toy model because pure $S = 1/2$ spins usually do not accommodate strong Ising anisotropy. However, it has become apparent that heavy 4d and 5d transition metal compounds exhibit correlated electron physics and may realize Kitaev behaviour, owing to the interplay between correlation and strong spin-orbit coupling³⁴. Promising hosts of the Kitaev QSL are Ir^{4+} oxides and a Ru^{3+} chloride with a d^5 electronic configuration and rock-salt-related honeycomb structure.

In Ir^{4+} oxides with octahedral coordination of oxygen ions, the oxygen octahedron produces a large crystal-field splitting of 3–4 eV, which wins over Hund's first rule favouring the high-spin state. All five d electrons are therefore accommodated in the triply degenerate t_{2g} manifold that comprises d_{xy} , d_{yz} and d_{zx} orbitals (FIG. 2a). Spin-orbit coupling λ_{SO} , relativistic in origin, is as large as 0.5 eV for heavy elements like Ir, much larger than the typical 0.1–0.2 eV crystal-field splitting of the t_{2g} manifold, which is produced by cubic-symmetry-breaking distortions of the oxygen octahedra. The effect of spin-orbit coupling can be introduced by treating the t_{2g} manifold as a triply degenerate manifold of p orbitals, with orbital angular momentum $l = 1$. Projecting the $l = 2$ angular momentum operator of atomic d levels onto the t_{2g} manifold yields an effective angular momentum of magnitude $l_{\text{eff}} = 1$ but with negative sign, $l = -l_{\text{eff}} = -1$ (REF. 35). Under the dominant spin-orbit coupling, the t_{2g} manifold splits into the higher $j_{\text{eff}} = 1/2$ Kramers doublet and the lower $j_{\text{eff}} = 3/2$ quartet, with a splitting energy of $3\lambda_{\text{SO}}/2$. With five t_{2g} electrons, the $j_{\text{eff}} = 3/2$ quartet is fully occupied, and the $j_{\text{eff}} = 1/2$ Kramers doublet accommodates one electron or, equivalently, one hole. Because there is only one hole in the t_{2g} manifold, it is often convenient to describe the electronic configuration using a hole picture (FIG. 2b). Upon introducing hopping between the neighbouring sites, a half-filled and relatively narrow band derived from the $j_{\text{eff}} = 1/2$ doublet is formed. A modest on-site Coulomb repulsion $U \approx 2$ eV can open a charge gap in the half-filled band produced by the strong spin-orbit coupling³⁶, and thus Ir^{4+} oxides are often weak spin-orbital Mott insulators with $j_{\text{eff}} = 1/2$ moments. Such a spin-orbital Mott insulating state was first identified in the layered perovskite Sr_2IrO_4 (REF. 22).

In the limit of strong Hund's coupling (the LS -coupling scheme), rather than strong spin-orbit coupling (the jj -coupling scheme described above), the five localized t_{2g} electrons form a six-fold degenerate manifold made of total spin moment $S = 1/2$ and effective angular momentum $L_{\text{eff}} = 1$. The manifold splits into the ground state $J_{\text{eff}} = 1/2$ Kramers doublet and the higher $J_{\text{eff}} = 3/2$ quartet at $3\lambda_{\text{SO}}/2$, owing to the spin-orbit coupling between S and L_{eff} . This insulating state with the $J_{\text{eff}} = 1/2$ moment determined from the LS -coupling scheme is essentially equivalent to the $j_{\text{eff}} = 1/2$ Mott insulator, because both have only one hole involved in the ground state. The difference between the two limits

appears when virtual charge fluctuations generate the Ir^{5+} , d^4 configuration with unquenched Hund's coupling. In this case, as discussed below, the Hund's coupling between two holes in the virtual d^4 configuration is essential for a correct description of the exchange interactions, and the LS -coupling picture might be more convenient. Because the energy scale of Hund's coupling is not very different from that of λ_{SO} for Ir^{4+} , the real situation in iridates is likely in between the jj -coupling and the LS -coupling limits. In the other part of this Review, we use the notation J_{eff} for simplicity. The same picture as for Ir^{4+} oxides holds for an Ru^{3+} chloride with five 4d electrons. Because a smaller $\lambda_{\text{SO}} \approx 0.1$ eV and a larger Hund's coupling of ~ 0.4 eV are expected for the Ru^{3+} ion with five 4d electrons, Ru chloride is better described by the LS -coupling scheme than are Ir^{4+} oxides.

In the Ir^{4+} oxides and the Ru^{3+} chlorides, the spin-orbit coupling is much larger than the typical exchange interactions among magnetic ions. The low-energy magnetism of these compounds is thus dominated by the $J_{\text{eff}} = 1/2$ moments. The spin-orbital entangled $J_{\text{eff}} = 1/2$ wavefunction is composed of the quantum superposition of d_{xy} , d_{yz} and d_{zx} orbitals with equal amplitudes but complex phases that describe orbital motion. The up ($\uparrow$) and down ($\downarrow$) spins reside on different orbital states: $|J_{\text{eff}} = 1/2, J_{\text{eff}}^z = +1/2\rangle = \frac{1}{\sqrt{3}}(|d_{xy}, \uparrow\rangle + |d_{yz}, \downarrow\rangle + i|d_{zx}, \downarrow\rangle)$ and $|J_{\text{eff}} = 1/2, J_{\text{eff}}^z = -1/2\rangle = -\frac{1}{\sqrt{3}}(|d_{xy}, \downarrow\rangle - |d_{yz}, \uparrow\rangle + i|d_{zx}, \uparrow\rangle)$. The $J_{\text{eff}} = 1/2$ state has an electron density distribution with a cubic shape (FIG. 2b) and hosts a magnetic moment of $1\mu_B$, like a free $S = 1/2$ spin. However, its gyromagnetic factor $g = -2$ is opposite that of a spin, which is a manifestation of the unquenched orbital moment L . The orbital component L causes $J_{\text{eff}} = 1/2$ moments and their exchange interactions to be extremely sensitive to the local crystalline fields and the bonding geometry^{23,37}.

When neighbouring IrO_6 octahedra share one of their corner oxygens to form 180° Ir–O–Ir bonds, the super-exchange interaction between the two adjacent $J_{\text{eff}} = 1/2$ moments on Ir^{4+} is dominated by the isotropic Heisenberg term, despite the strong spin-orbit coupling^{23,38,39}. The emergence of isotropy (pseudo-spin rotational symmetry) occurs because the electron hopping between neighbouring ions conserves not only the spin but also the orbital index and, therefore, the $J_{\text{eff}} = 1/2$ quantum number. However, when the two IrO_6 octahedra share one of their edges to form 90° Ir–O₂–Ir bonds (FIG. 2c), the oxygen-mediated hopping becomes orbital non-conserving, leading to exchange couplings with a discrete symmetry. A destructive quantum interference of the two Ir–O–Ir paths in the super-exchange process completely suppresses the conventional Heisenberg term between the neighbouring $J_{\text{eff}} = 1/2$ moments. Instead, an Ising ferromagnetic exchange $-KJ_{\text{eff},i}^z J_{\text{eff},j}^z$ with easy axes perpendicular to the Ir–O₂–Ir plane, emerges. The exchange arises from a combination of hopping to the neighbouring $J_{\text{eff}} = 3/2$ orbitals and the subsequent Hund's coupling, which favours parallel alignment of the real spins (FIG. 2d)²³. By replacing the spin-orbital entangled pseudo-spin J_{eff} with S , one arrives at the essential ingredients of the Kitaev model: bond-specific $S = 1/2$ Ising interactions $-K_\gamma S_i^\gamma S_j^\gamma$ with a γ -axis perpendicular to the Ir–O₂–Ir bond plane (FIG. 2e,f).

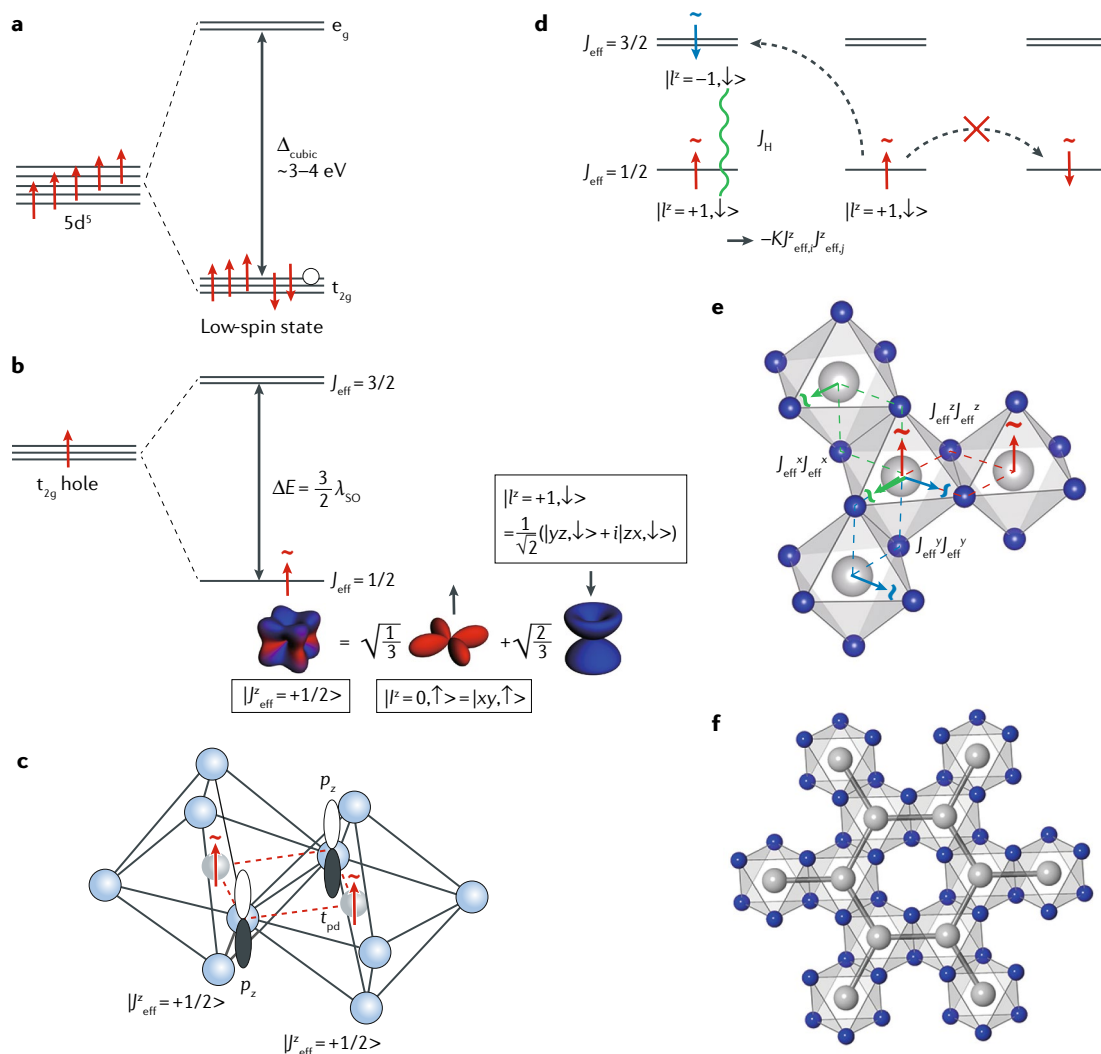


Fig. 2 | Local electronic states of octahedrally coordinated Ir^{4+} and Ru^{3+} ions and material realization of the Kitaev model. **a** | Splitting of five-fold degenerate d levels of d^5 Ir^{4+} and Ru^{3+} into the doublet e_g and the triplet t_{2g} due to the cubic crystal field Δ_{cubic} of O_6 or Cl_6 octahedra. The t_{2g} manifold consists of $|xy\rangle$, $|yz\rangle$ and $|zx\rangle$ orbitals. **b** | Spin-orbit coupling λ_{SO} further splits the t_{2g} manifold in panel **a** into the $J_{\text{eff}} = 1/2$ doublet and $J_{\text{eff}} = 3/2$ quartet with splitting energy $\Delta E = 3\lambda_{\text{SO}}/2$. In the hole picture shown, one hole is accommodated in the lower $J_{\text{eff}} = 1/2$ Kramers doublet. **c** | The edge-shared bond between the two IrO_6 or RuCl_6 octahedra gives rise to two 90° Ir–O–Ir or Ru–Cl–Ru bonds. The hole can hop (with amplitude t_{pd}) via the two oxygen p_z orbitals; hopping changes the orbital ‘colour’, that is, zx changes to yz and yz changes to zx . **d** | Quantum interference between the two hopping paths shown in panel **c** gives a selection rule $\Delta l^z = \pm 2$. As a result, hopping between the neighbouring $J_{\text{eff}} = 1/2$ orbitals is prohibited, and the conventional Heisenberg exchange is suppressed. Instead, hopping between the neighbouring $J_{\text{eff}} = 1/2$ and $J_{\text{eff}} = 3/2$ (more precisely, $J^z_{\text{eff}} = \pm 3/2$) orbitals dominates, and the Hund’s coupling J_{H} between real spins (green wavy line) leads to anisotropic Ising ferromagnetic coupling $-K J^z_{\text{eff},i} J^z_{\text{eff},j}$ between two $J_{\text{eff}} = 1/2$ pseudo-spins i and j , where K is the coupling strength. The Ising nature of this coupling follows from the hopping selection rules $\Delta l^z = \pm 2$ and $\Delta s^z = 0$, which dictate that a hole must return to the $J_{\text{eff}} = 1/2$ level with the same quantum numbers. **e** | The edge-shared bonds of an octahedron with its three neighbouring octahedra, accommodating Kitaev-type interactions. The easy axes of the $J_{\text{eff}} = 1/2$ pseudo-spin are perpendicular to the Ir–O₂–Ir or Ru–Cl₂–Ru bond planes and therefore orthogonal to each other for the three bonds. **f** | Honeycomb network of IrO_6 or RuCl_6 octahedra, commonly seen in Na_2IrO_3 , $\alpha\text{-Li}_2\text{IrO}_3$, $\alpha\text{-RuCl}_3$ and the other 2D honeycomb iridates in TABLE 1. The network comprising the three bonds in panel **e** can be viewed as a material realization of the Kitaev model in FIG. 1a. $\uparrow$, up-spin moment; $\downarrow$, down-spin moment; l^z , z-component of orbital angular momentum.

A_2IrO_3 ($\text{A} = \text{Na}, \text{Li}$) was first proposed for realization of the Kitaev model²³ and consists of alternating 2D layers of IrO_6 octahedra forming a honeycomb network (FIG. 2f) and Na or Li (FIG. 3a). Each IrO_6 octahedron shares its edges with the neighbouring IrO_6 octahedra and forms three 90° Ir–O₂–Ir bonds, with

the bonding plane orthogonal to the other two (FIG. 2f). The super-exchange process through Ir–O₂–Ir bonds gives rise to three kinds of Ising ferromagnetic bonds, with the bond-dependent easy axes orthogonal to each other. If the interaction between the neighbouring pseudo-spin $J_{\text{eff}} = 1/2$ moments is dominated by such

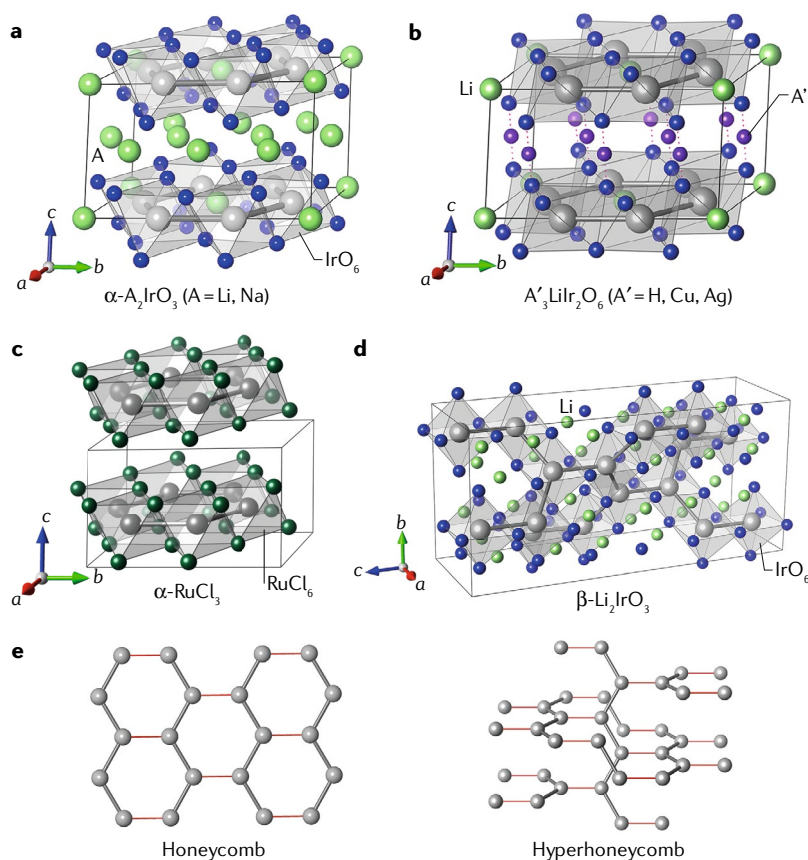


Fig. 3 | Kitaev candidate materials. **a** | The crystal structure of Na_2IrO_3 and $\alpha\text{-Li}_2\text{IrO}_3$. **b** | The crystal structure of $\text{A}'_3\text{LiIr}_2\text{O}_6$ (A' = H, Cu and Ag). **c** | The crystal structure of $\alpha\text{-RuCl}_3$. **d** | The crystal structure of $\beta\text{-Li}_2\text{IrO}_3$ with the hyperhoneycomb Ir sublattice shown in panel **e**. **e** | The relationship between the 2D honeycomb and the 3D hyperhoneycomb lattices. The bonds in the honeycomb lattice are decomposed into the zigzag chains (grey) and bridges (red). The crystal structures are visualized using VESTA software¹²⁹.

bond-dependent Ising interactions, the honeycomb network of A_2IrO_3 is a realization of the Kitaev model. The marriage of QSL physics and correlated electron physics kicked off the exploration of Kitaev candidate materials including Na_2IrO_3 (REF.⁴⁰). However, as discussed below, additional interactions always exist in materials and cannot be neglected.

Candidate materials. After the theoretical prediction of Kitaev physics in Na_2IrO_3 (REF.²³), many iridium and ruthenium compounds were recognized as candidate hosts of the Kitaev model (FIG. 3; TABLE 1). The Na^+ ions in Na_2IrO_3 can be replaced with Li^+ and Cu^+ ions^{41,42} (FIG. 3a). $\alpha\text{-Li}_2\text{IrO}_3$ has the same structure as Na_2IrO_3 and was investigated previously as a possible material for Li-ion batteries⁴³ and rediscovered as a Kitaev candidate⁴⁴. A second generation of 2D Kitaev candidates, $\text{H}_3\text{LiIr}_2\text{O}_6$, $\text{Cu}_3\text{LiIr}_2\text{O}_6$ and $\text{Ag}_3\text{LiIr}_2\text{O}_6$ (REFS^{45–49}), arise by applying an ion-exchange technique to $\alpha\text{-Li}_2\text{IrO}_3$ to replace the Li^+ ions sitting between the honeycomb layers with other monovalent ions. The chemical formula $\text{A}'_3\text{LiIr}_2\text{O}_6$ (A' = H, Cu and Ag) can be compared with $\alpha\text{-Li}_2\text{IrO}_3 = 1/2 (\text{Li}'_3\text{LiIr}_2\text{O}_6)$, where Li' represents Li ions between the honeycomb LiIr_2O_6 layers. The interlayer A' ions in $\text{A}'_3\text{LiIr}_2\text{O}_6$ form straight dumbbell

bonds with oxygens above and below, resulting in a different stacking pattern of LiIr_2O_6 layers from those of Na_2IrO_3 and $\alpha\text{-Li}_2\text{IrO}_3$ (REF.⁴⁷) (FIG. 3b). $\alpha\text{-RuCl}_3$ (FIG. 3c) comprises essentially the same honeycomb network of edge-sharing RuCl_6 octahedra and can be viewed as a fully Na-deficient analogue of Na_2IrO_3 (REFS^{50,51}).

$\beta\text{-Li}_2\text{IrO}_3$ (REF.⁵²) and $\gamma\text{-Li}_2\text{IrO}_3$ (REF.⁵³) were discovered as 3D analogues of $\alpha\text{-Li}_2\text{IrO}_3$. A 2D honeycomb lattice can be viewed as an assembly of 1D zigzag chains connected by bridging bonds, all confined in the same plane. The 3D honeycomb structure of $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$ can be derived from a 3D stack of such 2D honeycomb planes. In the Ir sublattice of $\beta\text{-Li}_2\text{IrO}_3$ (FIG. 3d), called a hyperhoneycomb, the zigzag chains in the 3D stack are rotated around the bridging bonds in an alternating pattern and reconnected to the zigzag chains in the layers above and below by the bridging bonds⁵² (FIG. 3e). In $\gamma\text{-Li}_2\text{IrO}_3$, which has a structure called a stripy honeycomb, the stripes with hexagons, consisting of two zigzag chains with bridging bonds in between, are rotated alternately around the stripe-bridging bonds and reconnected to the stripes in the layers above and below⁵³. In both $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$, all the Ir sites are crystallographically equivalent and remain octahedrally coordinated by oxygen atoms. As in $\alpha\text{-Li}_2\text{IrO}_3$, each IrO_6 octahedron shares its three orthogonal edges with the three neighbouring IrO_6 octahedra, forming three bonds. The local bonding configurations of $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$ are the same as that of $\alpha\text{-Li}_2\text{IrO}_3$, leading to the same competition between three Ising bonds with orthogonal easy axes. The 3D Kitaev model on the hyperhoneycomb ($\beta\text{-Li}_2\text{IrO}_3$ -type) lattice was investigated even before the compounds were discovered and shown to have a QSL as its ground state⁵⁴. These 3D honeycomb iridates represent the realization of such 3D Kitaev models.

In all candidates, the IrO_6 or RuCl_6 octahedra are not regular but have small non-cubic distortions^{41,47,51,55}. In the case of the 2D honeycomb compounds, compression perpendicular to the layer direction arises because the two O_3 or Cl_3 triangles above and below Ir or Ru, facing each other and forming the O_6 or Cl_6 octahedron, come closer to each other. As a result, the Ir–O–Ir or Ru–Cl–Ru angle becomes more than 90° , and a trigonal crystal field is produced. This very likely reduces the Kitaev-type interaction and enhances some of the other interactions. In addition, the interlayer coupling of those layered compounds is typically very weak. All the samples reported to date suffer from the presence of stacking faults, seen as a broadening of X-ray diffraction peaks^{47,48}. In the case of $\alpha\text{-RuCl}_3$, the refined crystal structures and, as discussed below, the magnetic ordering transitions were reported to show sample dependence⁵¹.

Single crystals of Na_2IrO_3 can be grown by a flux method⁴⁰. The single-crystal growth of $\alpha\text{-Li}_2\text{IrO}_3$ is not as easy as that of Na_2IrO_3 , partly because of the existence of β -type and γ -type polymorphs^{52,53}. $\alpha\text{-Li}_2\text{IrO}_3$ single crystals can be grown by utilizing a vapour transport technique⁵⁶, while single crystals of $\alpha\text{-RuCl}_3$ can be obtained by a Bridgman technique⁵⁷, vacuum sublimation^{50,58} or vapour transport technique⁵¹. The second generation of Kitaev materials $\text{A}'_3\text{LiIr}_2\text{O}_6$ (A' = Ag, Cu, H)^{45–49} and

Table 1 | Representative Kitaev candidate materials and a summary of their physical properties

Materials	Crystal structure (space group)	T_{mag}	Anisotropy	p_{eff} (μ_B)	θ_{CW} (K)	Magnetic ground state	Refs
Na_2IrO_3	2D (C2/m)	15 K	$\chi_c > \chi_{ab}$	1.81 (ab) 1.94 (c)	−176 (θ_{ab}) −40 (θ_c)	Zigzag	40,60,68,69
$\alpha\text{-Li}_2\text{IrO}_3$	2D (C2/m)	15 K	$\chi_{ab} > \chi_c$	1.50 (ab) 1.58 (c)	+5 (θ_{ab}), −250 (θ_c)	Spiral	44,56,71
$\text{H}_3\text{LiIr}_2\text{O}_6$	2D (C2/m)	–	$\chi_{ab} > \chi_c$	1.60	−105	Spin-liquid	46
Cu_2IrO_3	2D (C2/c)	2.7 K	Not known	1.93 (1)	−110	Antiferromagnetic order or spin-glass	42
$\text{Cu}_3\text{LiIr}_2\text{O}_6$	2D (C2/c)	15 K	Not known	2.1 (1)	−145	Antiferromagnetic order	49
$\text{Ag}_3\text{LiIr}_2\text{O}_6$	2D (R-3m ^a)	~12 K	Not known	1.77		Antiferromagnetic order	48
$\alpha\text{-RuCl}_3$	2D (C2/m or P3 ₁ 12, or R-3); T and sample dependent	7 K and/or 14 K (see text)	$\chi_{ab} > \chi_c$	2.33 (ab), 2.71 (c)	+39.6 (θ_{ab}), −216.4 (θ_c)	Zigzag	51,58,67, 70,130
$\beta\text{-Li}_2\text{IrO}_3$	3D (Fddd)	38 K	$\chi_b > \chi_c > \chi_a$	1.87 (a) 1.80 (b) 1.97 (c)	−90.2 (θ_a) +12.9 (θ_b) +21.6 (θ_c)	Spiral	52,72,93
$\gamma\text{-Li}_2\text{IrO}_3$	3D (Cccm)	39.5 K	$\chi_b > \chi_c > \chi_a$	~1.6	+40	Spiral	53,73

^aR-3m assumed in REF.⁴⁸ because of strong stacking disorder.

Cu_2IrO_3 (REF.⁴²) can be synthesized using a soft-chemical ion-exchange reaction, by soaking powder of $\alpha\text{-Li}_2\text{IrO}_3$ (or Na_2IrO_3 for Cu_2IrO_3) in a molten salt or aqueous solution containing A' ions. Only polycrystalline powder is available for the second-generation materials at the time of writing. The single-crystal growth of these materials is challenging but crucially important for further investigations.

Behaviour of Kitaev candidates

Electronic properties. All the Kitaev candidate materials listed in TABLE 1 exhibit an insulating behaviour and a localized moment magnetism, indicating that they are $J_{\text{eff}} = 1/2$ spin-orbital Mott insulators. The optical conductivity spectrum for Na_2IrO_3 shows a sizeable charge gap of ~300 meV (REF.⁵⁹), which is consistent with an activation energy in the resistivity of ~100 meV at room temperature^{40,60}. Transport activation energies comparable to that of Na_2IrO_3 have been observed in the other honeycomb-based iridium oxides^{44,52}, implying similar charge gaps. The relatively small size of the charge gap suggests that they form weak Mott insulators owing to the moderate Coulomb repulsion $U \approx 2$ eV. For $\alpha\text{-RuCl}_3$, optical conductivity measurements show a charge gap of ~1 eV, substantially larger than that of Na_2IrO_3 , likely reflecting a larger U for 4d Ru and a narrow bandwidth originating from the strong ionicity of chloride⁶¹.

In the resonant inelastic X-ray scattering spectra at the Ir L_3 ($2p_{3/2} \rightarrow 5d$) edge for Na_2IrO_3 , $\alpha\text{-Li}_2\text{IrO}_3$ and $\beta\text{-Li}_2\text{IrO}_3$, the presence of low-energy excitations of ~0.7 eV is commonly observed^{62,63}, which corresponds to the excitation energy of $3\lambda_{\text{SO}}/2$ from $J_{\text{eff}} = 1/2$ to $3/2$ within the t_{2g} manifold, and indicates a spin-orbit coupling $\lambda_{\text{SO}} \approx 0.4\text{--}0.5$ eV. A splitting of the 0.7 eV peak is observed⁶², likely originating from the splitting of $J_{\text{eff}} = 3/2$ states due to the

trigonal lattice distortion of IrO_6 octahedron. The splitting is smaller than $3\lambda_{\text{SO}}/2 \approx 0.7$ eV, meaning that the $J_{\text{eff}} = 1/2$ picture is valid as a first approximation, but the effect of the trigonal crystal field may not be negligibly small.

The presence of excitations between $J_{\text{eff}} = 1/2$ and $J_{\text{eff}} = 3/2$ in analogues to the iridates was identified by INS in $\alpha\text{-RuCl}_3$ (REF.⁶⁴) and is consistent with the expected λ_{SO} of 0.1 eV for 4d Ru. Despite the small spin-orbit coupling, the $J_{\text{eff}} = 1/2$ picture holds for $\alpha\text{-RuCl}_3$ and is likely an even better approximation than that in the iridium oxides. The RuCl_6 octahedron is less trigonally distorted than the IrO_6 octahedra in the iridium oxides, which reduces the crystal-field splitting to a smaller value than the spin-orbit coupling of $\lambda_{\text{SO}} \approx 0.1$ eV (REF.⁶⁵).

$J_{\text{eff}} = 1/2$ magnetism. A numerical calculation of magnetic susceptibility $\chi(T)$ for the Kitaev model with uniform ferromagnetic couplings $K = K_x = K_y = K_z$ indicates an isotropic Curie–Weiss behaviour at high temperatures, with a Curie constant for $S = 1/2$ and $g = 2$ and with a ferromagnetic Curie–Weiss temperature $\theta_{\text{CW}} = K/4$. Upon lowering T below $\sim K/k_B$, $\chi(T)$ deviates downwards from the Curie–Weiss behaviour and crosses over to almost T -independent behaviour around a characteristic temperature T_H below which the spin–spin correlations saturate⁶⁶.

The magnetic susceptibilities $\chi(T)$ of the Kitaev candidate compounds^{40,67} are summarized in FIG. 4a,b. The relevant magnetic parameters estimated from $\chi(T)$ in FIG. 4 are listed in TABLE 1. The Curie–Weiss behaviour at high temperatures is indicative of the localized moment magnetism. The effective moments, determined from the slope of the linear behaviour of $\chi^{-1}(T)$, are close to $p_{\text{eff}} = 1.73 \mu_B$ expected for the pure $J_{\text{eff}} = 1/2$ state, and equivalent to the case of $S = 1/2$ and $g = 2$,

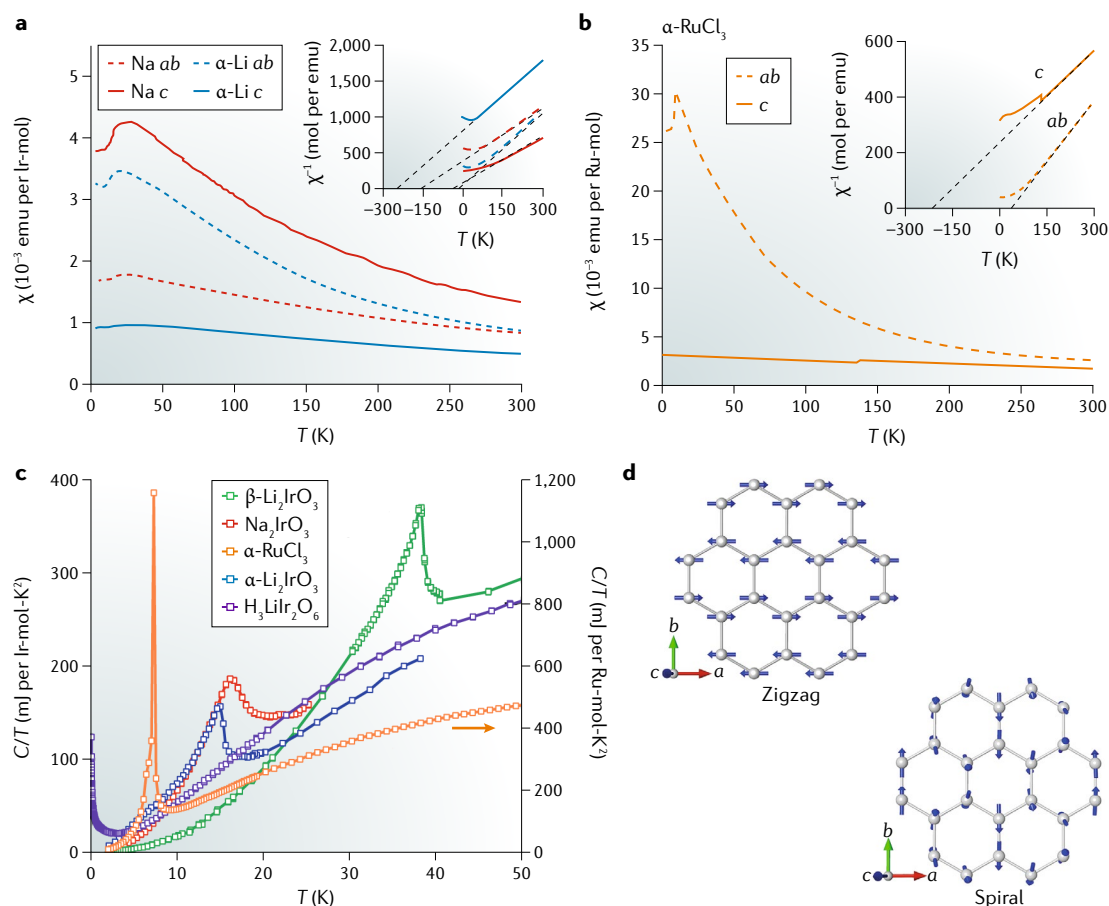


Fig. 4 | Magnetic ordering in Kitaev candidate materials. **a** | Magnetic susceptibility $\chi(T)$ as a function of temperature T for 2D honeycomb iridate Na_2IrO_3 (Na)⁴⁰ and $\alpha\text{-Li}_2\text{IrO}_3$ ($\alpha\text{-Li}$) with magnetic fields parallel (ab) and perpendicular (c) to the honeycomb plane. The inset shows the Curie–Weiss plot, $\chi^{-1}(T)$, of data in the main panel. **b** | $\chi(T)$ for 2D honeycomb ruthenium compound $\alpha\text{-RuCl}_3$ with magnetic fields parallel (ab) and perpendicular (c) to the honeycomb plane⁶⁷. The inset shows the Curie–Weiss plot of data in the main panel. **c** | The low-temperature specific heat $C(T)$ for Na_2IrO_3 , $\alpha\text{-Li}_2\text{IrO}_3$ (REF.⁵⁶), $\text{H}_3\text{LiIr}_2\text{O}_6$ (REF.⁴⁶), $\alpha\text{-RuCl}_3$ (REF.⁵¹) and $\beta\text{-Li}_2\text{IrO}_3$. All materials show an anomaly in $C(T)$, indicative of an ordering transition, except $\text{H}_3\text{LiIr}_2\text{O}_6$. **d** | Zigzag and spiral spin-ordering patterns for Na_2IrO_3 and $\alpha\text{-Li}_2\text{IrO}_3$.

and are material independent (TABLE 1). This is consistent with all the compounds in TABLE 1 being $J_{\text{eff}} = 1/2$ Mott insulators. A sizeable field-orientation-dependent anisotropy is present in $\chi(T)$, suggesting the presence of bond-dependent anisotropic magnetic couplings and a trigonal field effect. The deviation from the Curie–Weiss behaviour occurs at temperatures between 100 K and 200 K, indicating an approximate energy scale of magnetic interactions of 100–200 K. The Curie–Weiss temperatures θ_{CW} are negative (that is, antiferromagnetic), except for $\alpha\text{-RuCl}_3$ ($B_{\parallel}$ plane) and $\beta\text{-Li}_2\text{IrO}_3$ ($B_{\parallel}$ b axis or c axis), and depend strongly on the materials and the field orientations, ranging from -200 K to almost zero. It is clear that the Kitaev-type ferromagnetic coupling is not the only important interaction and that appreciable antiferromagnetic interactions are present. The near-zero θ_{CW} despite the deviation from the Curie–Weiss behaviour above 100 K indicates a cancellation of antiferromagnetic and ferromagnetic interactions, evidencing the presence of more than one kind of interaction in these compounds. The wide range of θ_{CW} values very likely mirrors the sensitivity of the interactions

to the local lattice structure and the details of chemical bonding, arising from the spin-orbital entanglement.

Long-range order. Except for $\text{H}_3\text{LiIr}_2\text{O}_6$, all the candidate compounds show a clear signature of long-range magnetic ordering rather than a QSL state, which is evidenced by a kink in $\chi(T)$ (FIG. 4a,b; TABLE 1) and the anomaly in specific heat $C(T)$ ^{40,51,56} (FIG. 4c). The ordering temperature T_{mag} is one order of magnitude lower than the magnetic interaction energy scale of 100–200 K inferred from the deviation from Curie–Weiss behaviour. It indicates the presence of magnetic frustration, which is consistent with the candidates being close to a frustrated Kitaev QSL. In accord with the presence of frustration, the entropy change around T_{mag} is only 10–30% of $R\ln 2$ (REFS^{40,51,52,56}), that is, 10–30% of the full entropy of the $J_{\text{eff}} = 1/2$ doublet.

Resonant X-ray diffraction and neutron measurements indicate that below T_{mag} , Na_2IrO_3 (REFS^{58,69}) and $\alpha\text{-RuCl}_3$ (REFS^{51,58,70}) form a zigzag ordering in which ferromagnetic zigzag chains are coupled antiferromagnetically across the bridging bonds. $\alpha\text{-Li}_2\text{IrO}_3$ shows

Box 1 | Non-Kitaev interactions and induced magnetically ordered phases

The generic nearest-neighbour exchange Hamiltonian for the undistorted layered hexagonal $A_2\text{IrO}_3$ and $\alpha\text{-RuCl}_3$ compounds includes three symmetry-allowed terms^{23,77–81,126}

$$\mathcal{H} = \sum_{\langle ij \rangle_\gamma} \left\{ -KS_i^\gamma S_j^\gamma + I(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha) + JS_i \cdot S_j \right\} \quad (3)$$

Here, $\langle ij \rangle_\gamma$ denotes the three inequivalent nearest-neighbour bonds of the honeycomb lattice, with $\gamma = x, y$ or z , and $S_i^{\alpha(\beta)}$ and S_i^γ stand for the in-plane and out-of-plane Cartesian components of the $S = 1/2$ pseudo-spins $\mathbf{S}$ with respect to the $\text{Ir-O}_2\text{-Ir}$ plane, respectively. $\mathbf{S}$ represents the $J_{\text{eff}} = 1/2$ Kramers pairs.

The first Kitaev coupling K originates from the combined effects of the anion-mediated super-exchange and Hund's coupling, as discussed in the main text. All available theoretical methods, including perturbation theory, ab initio calculations and quantum chemistry calculations, suggest that the Kitaev term dominates the other two terms and is ferromagnetic ($K > 0$)^{23,60,79}. The off-diagonal symmetric anisotropy, the I term, is of next-to-leading order and requires both direct d–d and anion-mediated d–p electron transfer⁷⁸. The isotropic Heisenberg coupling J is antiferromagnetic ($J > 0$). It is the smallest term because it predominantly originates from the weak direct d–d hybridization.

This minimal model has been studied using analytical and numerical techniques, and the rich phase behaviour is now understood^{77,78,116,127,128}. For $I = 0$, the model supports four magnetically ordered phases with collinear spin patterns of ferromagnetic, antiferromagnetic, stripy and zigzag types^{77,116}. Most remarkably, the phase diagram includes a finite stability window for the quantum spin liquid (QSL) phase close to the Kitaev limit ($J = 0$). Finite I further enriches the phase diagram by adding non-collinear and incommensurate spiral phases⁷⁸.

The experimentally detected zigzag magnetic ordering in Na_2IrO_3 and $\alpha\text{-RuCl}_3$ appears on the theoretical phase diagram of the minimal nearest-neighbour model for antiferromagnetic Kitaev coupling ($K < 0$). However, it is widely believed that the Kitaev term in these compounds is of ferromagnetic type instead. It has therefore been proposed that exchange couplings beyond nearest-neighbour are responsible for stabilizing the zigzag order^{55,82} observed in Na_2IrO_3 and $\alpha\text{-RuCl}_3$. Indeed, the ab initio parametrization of further-neighbour interactions indicate that they become sizeable in these compounds⁶⁰.

The non-cubic crystal field arising from the distorted octahedra mixes $J_{\text{eff}} = 1/2$ and $J_{\text{eff}} = 3/2$ states. The ground state remains doubly degenerate, protected by time-reversal symmetry, and can still be described by pseudo-spin $J_{\text{eff}} = 1/2$. However, because the crystal field modifies the associated wavefunction, the abovementioned destructive quantum interference is no longer exact, and the isotropic Heisenberg term becomes larger. Distortion-induced lowering of the crystal symmetry also gives rise to other non-Kitaev interactions, such as an antisymmetric anisotropy known as the Dzyaloshinskii–Moriya term.

The perturbative effect of these non-Kitaev interactions in the QSL state was studied, revealing the drastic change in low-energy spin dynamics¹⁰³. Additional interactions could close a spin gap inherent to the Kitaev model, giving rise to a finite low-energy spectral weight in the dynamic spin structure factor. The magnetic-field-induced Majorana gap in the presence of non-Kitaev terms, for example, scales linearly with the applied field, instead of the cubic dependence $\Delta_M \approx H_x H_y H_z / K^2$ in the pure Kitaev limit¹⁰³.

a spiral ordering below T_{mag} along the zigzag chains⁷¹ (FIG. 4d). The 3D honeycomb iridates, $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$ show a complex incommensurate spiral order below $T_{\text{mag}} \approx 38$ K (REFS^{72,73}). The parallel behaviour of polymorphic $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$, including indistinguishably close magnetic ordering temperatures, has been suggested to be a result of the identical local connectivity of the three Ir bonds⁷⁴. The details of the magnetic ordering transition in $\alpha\text{-RuCl}_3$ are sample dependent: crystals with minimal stacking faults generally show a sharp transition at $T_{\text{mag}} \approx 7$ K (REF.⁵¹), corresponding to zigzag order in the plane with a 3-layer periodicity. Samples with significant presence of stacking faults show an additional broad transition near 14 K corresponding to a 2-layer periodicity of the zigzag

structure, and powders show only the broad 14 K transition. It is possible that other periodic layerings exist, corresponding to different values of T_{mag} .

The emergence of long-range magnetic ordering rather than a QSL state is believed to originate from the presence of additional interactions beyond the Kitaev model. Although the dominance of Kitaev interactions has been shown, for example, from the azimuthal angle dependence of resonant X-ray diffuse magnetic scattering in Na_2IrO_3 (REF.⁷⁵) and the dispersion of spin excitations obtained from INS in $\alpha\text{-RuCl}_3$ (REFS^{64,76}), there are other types of exchange interactions, such as Heisenberg exchange through direct overlap of d orbitals⁷⁷ and off-diagonal exchanges such as the $S_i^\alpha S_j^\gamma$ term^{78–80}, which bring the real materials away from the pure Kitaev model regime. Lattice distortions, such as compression of the IrO_6 octahedron perpendicular to the plane, give rise to an increased Ir–O–Ir bond angle from 90° and admixture of $J_{\text{eff}} = 3/2$ states with $J_{\text{eff}} = 1/2$ states^{79,80}, which may add extra exchange paths⁸¹. Further-neighbour interactions could also be relevant^{44,55,82–84}. The Kitaev–Heisenberg model with additional nearest-neighbour Heisenberg interactions was the first to be discussed in the literature⁷⁷. It gives a Kitaev spin liquid in the pure Kitaev limit, Néel order in the strong Heisenberg limit and stripy order in between⁷⁷. The experimentally observed zigzag and spiral orderings are not contained in the theoretical phase diagram of the ferromagnetic Kitaev–Heisenberg model. This has led to the discussion of the off-diagonal and further-neighbour interactions, which, for a certain range of parameters, reproduce the zigzag and incommensurate spiral orderings^{78–81,85,86} (see BOX 1 for details).

Towards quantum spin liquid

Phase control. Despite low-temperature magnetic ordering, the candidate materials for the Kitaev spin liquid have sizeable Kitaev interactions and may be tunable into a spin-liquid ground state by external perturbations (FIG. 5). The results of numerical calculations for an extended Kitaev model with Heisenberg and off-diagonal interactions (BOX 1) indeed indicate the presence of a QSL for a finite region of the parameter space^{78–80}. The suppression of magnetic ordering has been attempted by applying a magnetic field or high pressure.

$\alpha\text{-RuCl}_3$ undergoes a zigzag-type magnetic ordering at $T_{\text{mag}} \approx 7$ K, which was found to be suppressed dramatically by applying an in-plane magnetic field of $B_c \approx 7\text{--}8$ T (REFS^{58,87–89}) (FIG. 5a). Above B_c , the system shows no magnetic ordering down to well below 1 K. Around B_c , an induced moment of $\sim 0.6 \mu_B$ is observed⁵⁸, suggesting that a substantial portion of magnetic entropy is lifted. Nevertheless, unusual magnetic excitations and thermal transport were discovered in the field-induced critical paramagnet^{89–92}, which were discussed in connection with a Kitaev QSL and are attracting considerable attention from the community. This is described in detail below.

Similar field-induced suppression of the magnetic order is observed in 3D honeycomb $\beta\text{-Li}_2\text{IrO}_3$ and $\gamma\text{-Li}_2\text{IrO}_3$. By applying a magnetic field along the

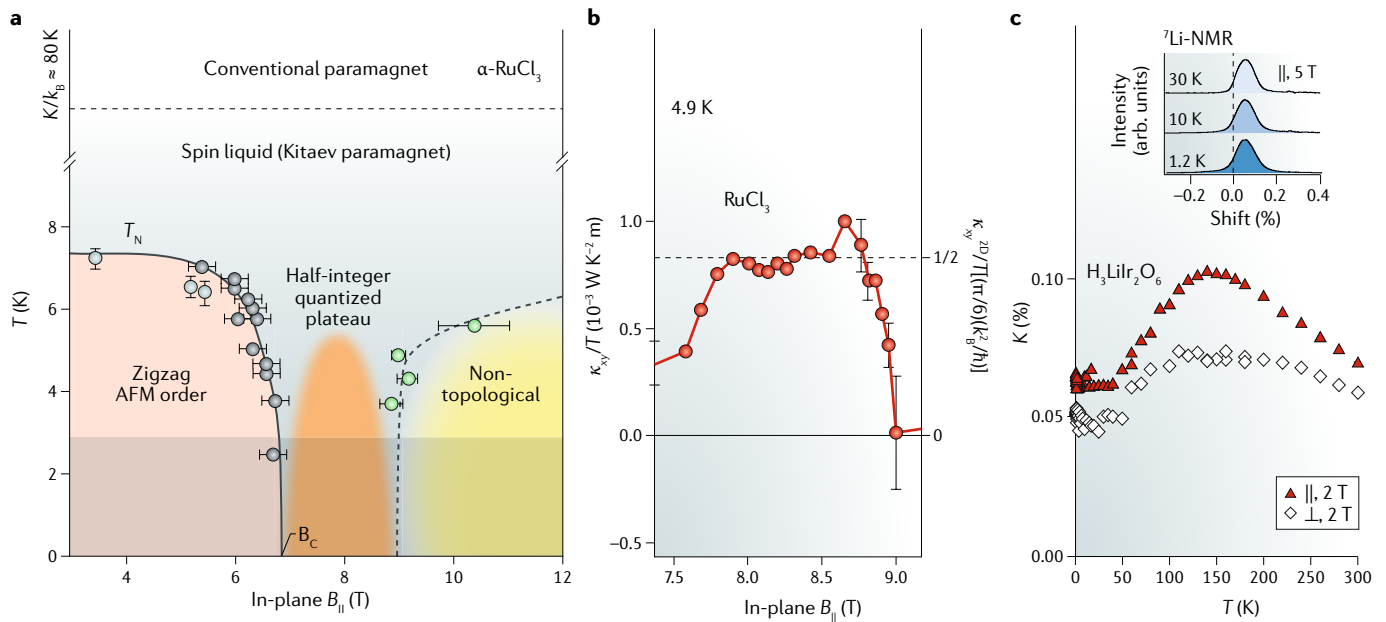


Fig. 5 | Magnetic-field-induced paramagnetism and quantum spin liquid behaviour in Kitaev candidate materials. **a** | Proposed phase diagram of α - RuCl_3 as a function of the in-plane magnetic field $B_{\parallel}$ (REF.⁸⁹). The zigzag antiferromagnetic (AFM) order is suppressed completely at approximately $B_c = 7\text{--}8$ T. Just above B_c , a topological phase with quantized thermal Hall conductivity emerges, followed by a nontopological phase. At temperatures above $K/k_B \approx 80$ K, the system is a conventional paramagnet. **b** | Quantized thermal Hall conductivity κ_{xy} as a function of in-plane $B_{\parallel}$ measured under 60° tilted magnetic field. The dotted line indicates a half-quantized value $\kappa_{xy}/T = 1/2(\pi/6)(k_B^2/\hbar)^2$. **c** | Temperature-dependent Knight shift $K_s(T)$ with magnetic field parallel ($\parallel$) and perpendicular ($\perp$) to the honeycomb plane. The inset shows ^7Li -NMR spectra at different temperatures, showing no broadening down to a low temperature. $\hbar$, Planck constant; K , Kitaev interaction; k_B , Boltzmann constant. Panels **a** and **b** are adapted from REF.⁸⁹, Springer Nature Limited. Panel **c** is adapted from REF.⁴⁶, Springer Nature Limited.

crystallographic b axis, which is the expected Ising axis of bridging bonds, the magnetic transition is smeared out and disappears above $B_c \approx 3$ T, at which a magnetic moment of $0.4 \mu_B$ is observed^{52,53,93,94}. This field-induced paramagnetic phase of 3D honeycomb compounds is much less explored than that of α - RuCl_3 .

Pressure is another external perturbation that controls magnetic interactions and thus the magnetic ground state. In α - Li_2IrO_3 , α - RuCl_3 and 3D β - Li_2IrO_3 , a first-order transition to a spin-singlet dimerized state under a pressure is commonly observed^{63,95–99}, in which a modulation of zigzag chains, the alternating contraction and elongation of Ir–Ir or Ru–Ru bonds, takes place. The Ir–Ir distance of contracted bonds is even shorter than that of metallic Ir, suggesting the formation of strong Ir_2 -dimer molecules rather than a weak spin-Peierls-like transition^{63,98}. The strong dimerization appears to compete with the $J_{\text{eff}} = 1/2$ Mott state that has isotropic superposition of the three orbital states, because a specific orbital state along the dimer molecule is selected to form the dimerized bond. Similar strong dimerization occurs in many 3d and 4d transition metal oxides, including Li_2RuO_3 with the same honeycomb structure¹⁰⁰.

Quantum spin liquid state in $\text{H}_3\text{LiIr}_2\text{O}_6$. Another approach to control magnetic interactions in Kitaev candidates is chemical modification. The second generation of 2D honeycomb iridates, $\text{A}'_3\text{LiIr}_2\text{O}_6$ ($\text{A}' = \text{H, Cu and Ag}$)^{45–49}, is a typical playground for such an approach. A QSL state was recently discovered in protonated

$\text{H}_3\text{LiIr}_2\text{O}_6$ (REF.⁴⁶). The honeycomb layers of $\text{H}_3\text{LiIr}_2\text{O}_6$ are more compressed perpendicular to the plane than the first generation of 2D honeycomb iridates^{45,47}, and this compression could modify the magnetic interactions appreciably. The evidence for the absence of both short-range and long-range magnetic ordering can be seen in the ^7Li -NMR and ^1H -NMR spectra, which do not show any broadening down to below 1 K (FIG. 5c). The Knight shift $K_s(T)$ obtained from the spectra, as well as $\chi(T)$, shows a Curie–Weiss behaviour with $\theta_{\text{CW}} \approx -100$ K at high temperatures, analogous to those of the first-generation candidates. As the temperature is lowered below 150 K, $K_s(T)$ of $\text{H}_3\text{LiIr}_2\text{O}_6$ shows a broad peak around 130 K, in contrast to the first-generation candidates, and decreases to a finite value in the $T = 0$ limit (FIG. 5c).

Signatures of local, particle-hole-symmetric, low-energy fermionic excitations are observed in the magnetization, NMR relaxation and specific heat at low temperatures, originating from a small number of spin defects. The defect contributions to these quantities follow a scaling with B/T , and although no theoretical consensus has emerged, it has been suggested that the scaling could arise from spin defects embedded in the bulk QSL forming random singlets¹⁰¹ or from a peculiar band dispersion of Majorana fermions produced by an interlayer coupling¹⁰². After subtracting the scaled contribution originating from the defects, only a field-independent T^3 contribution to the specific heat is observed below $T = 5$ K, which very likely

originates from the lattice. The absence of appreciable magnetic entropy at low temperatures may suggest the presence of a gap in the spin excitations. The observed behaviour is distinct from that expected for the ‘pure’ Kitaev QSL, which has a specific heat anomaly with an entropy of 50% of $R\ln 2$ associated with ordering the Z_2 fluxes (localized Majorana fermions)³². It was argued that the presence of non-Kitaev interactions may change the nature of the QSL appreciably from that in the pure Kitaev limit¹⁰³. For $\text{H}_3\text{LiIr}_2\text{O}_6$, the randomness of the H positions was also discussed as playing an important role in stabilizing the QSL state^{104,105}. Unveiling how this relates to Kitaev physics is an interesting challenge for the near future.

Fractional magnetic excitations

Spectroscopic measurements. As discussed above, one defining characteristic of the QSL is the presence of fractional magnetic excitations (FIG. 6), and the predicted Majorana fermions in the Kitaev QSL are of widespread interest. A variety of spectroscopic techniques have been used to examine magnetic excitations in iridates and, in particular, $\alpha\text{-RuCl}_3$, including INS^{55,64,76,92,106,107}, Raman spectroscopy^{108,109}, electron spin resonance (ESR)¹¹⁰, NMR^{90,111,112} and THz spectroscopy^{91,113–115}.

With the exception of $\text{H}_3\text{LiIr}_2\text{O}_6$ (REF.⁴⁶), the Kitaev candidate materials discussed in this Review order magnetically at low temperature, albeit with reduced ordered moments. In principle, the magnetically ordered states support conventional spin-wave modes. In collinear magnets without frustrated or competing interactions, excitations have $S = 1$, and they typically appear as peaks that are sharp in energy for a given momentum, that is, as magnons with well-defined dispersions. A single-crystal measurement of the spin-wave dispersions and intensities via INS often enables the direct determination of the effective spin Hamiltonian parameters.

In Kitaev candidates, the spin-wave spectra themselves can be calculated using parameters proposed to describe $\alpha\text{-RuCl}_3$ from theoretical considerations^{116,117}. There is, however, more than one set of parameters that can reproduce the experimental results reasonably, likely because of strong frustration. The iridate materials present additional challenges because Ir is a strong neutron-absorber and the magnetic form factor appearing in the neutron scattering cross section¹¹⁸ decays rapidly as the magnitude of the momentum transfer is increased. Moreover, single crystals suitable for INS are not necessarily available, requiring measurements to be made using polycrystalline material. Despite these difficulties, INS measurements of Na_2IrO_3 powder⁵⁵ proved very informative. For example, the shape of the scattering threshold alone established the magnetic order as zigzag, not stripy (BOX 2). Similar measurements on Ru-based materials such as $\alpha\text{-RuCl}_3$ are generally much easier. In $\alpha\text{-RuCl}_3$ powders, the shape of the low-energy, low-momentum scattering threshold is well defined, and an additional high-energy feature is detected with a temperature dependence that is incompatible with simple spin-wave theory⁶⁴ (BOX 2). The full significance of the inelastic scattering data is more apparent in single-crystal measurements, as discussed below.

INS data from single crystals of $\alpha\text{-RuCl}_3$ have been measured using time-of-flight methods^{76,92} (FIG. 6a,b). The scattering associated with the zone centre (Γ point) of the honeycomb lattice (FIG. 6a) shows that in the ordered state, the response has two sharp peaks superposed on a broad continuum that continues to high energies. The sharp peaks arise from spin waves in the ordered state and disappear above T_N , leaving only the continuum. The continuum is temperature independent up to well above 100 K. The momentum dependence of the scattering at $T = 2$ K (FIG. 6b) shows the spin waves as low-energy features with minima at the wavevectors $(\pm\frac{1}{2}, 0, 0)$, and the continuum, centred at the Γ point, is broad in both energy and momentum. The spin-wave spectrum is gapped in the ordered state; presumably, this is a signature of anisotropy in the interactions. The magnetic order in $\alpha\text{-RuCl}_3$ is suppressed by the application of an external magnetic field with magnitude $B_c = 7\text{--}8$ T along the in-plane $\langle 110 \rangle$ direction in reciprocal space (trigonal description) (FIG. 5a). In the resulting disordered state, the spin waves are again suppressed (FIG. 6b), whereas the continuum gains intensity at low energies. Although the energy gap at the Γ point softens as the field magnitude approaches B_c , it appears that the continuum scattering is gapped above B_c ; this can also be seen in scattering intensity as a function of energy⁹². We note that a gap in the disordered phase is thought to be required to observe a quantized thermal Hall effect²¹. The magnetic interactions in $\alpha\text{-RuCl}_3$ are 2D, and the scattering data shown in FIG. 6b are integrated over a range of out-of-plane momentum values under the assumption that this enhances the counting statistics with no significant loss of information. However, it must be kept in mind that the ordered state itself is 3D and, by corollary, so is the spin-wave spectrum.

The coexistence of sharp spin waves and broad continuum scattering in the ordered state is reminiscent of the situation in systems of weakly coupled $S = 1/2$ Heisenberg antiferromagnetic chains¹¹⁹. The natural excitations of the 1D system are fractionalized spinons that lead to a scattering continuum. At low temperatures, the coupled chains order, and the lowest-energy excitations are 3D spin waves; these coexist with the higher-energy spinon spectrum. Above T_N , the spin waves disappear, but the spinon scattering remains. This behaviour is experimentally verifiable, for example, in the quasi-1D magnet KCuF_3 (REF.¹²⁰).

It is interesting to consider whether an analogous situation exists in $\alpha\text{-RuCl}_3$. The exact response function of a pure isotropic Kitaev model²⁹ shows a sharp mode at very low energies corresponding to a static gauge flux and a weak continuum at higher energies arising from mobile Majorana fermions. Perturbations away from this limit¹²¹ result in a broadening of the sharp peak (FIG. 7a,b). In the ordered state of $\alpha\text{-RuCl}_3$, the low-energy part of the spectrum is dominated by spin waves; however, the overall extent and form of continuum scattering at high energies compares well to the calculated response functions, consistent with the expectations for a system proximate to a Kitaev QSL. Moreover, the temperature dependence of the continuum scattering is also consistent

with expectations for Majorana fermions in the Kitaev QSL^{55,64,66,106}. However, additional work is required to arrive at a definitive understanding of the INS measurements; at the time of writing, there is still some disagreement concerning the correct Hamiltonian describing

α -RuCl₃ (REFS^{116,122}). The response function, calculated by exact diagonalization of a finite-sized system at $T = 0$, can exhibit broad intensity distributions (FIG. 7c), inducing some to argue that the scattering is better described by unstable magnons¹²³. Given the similarity of

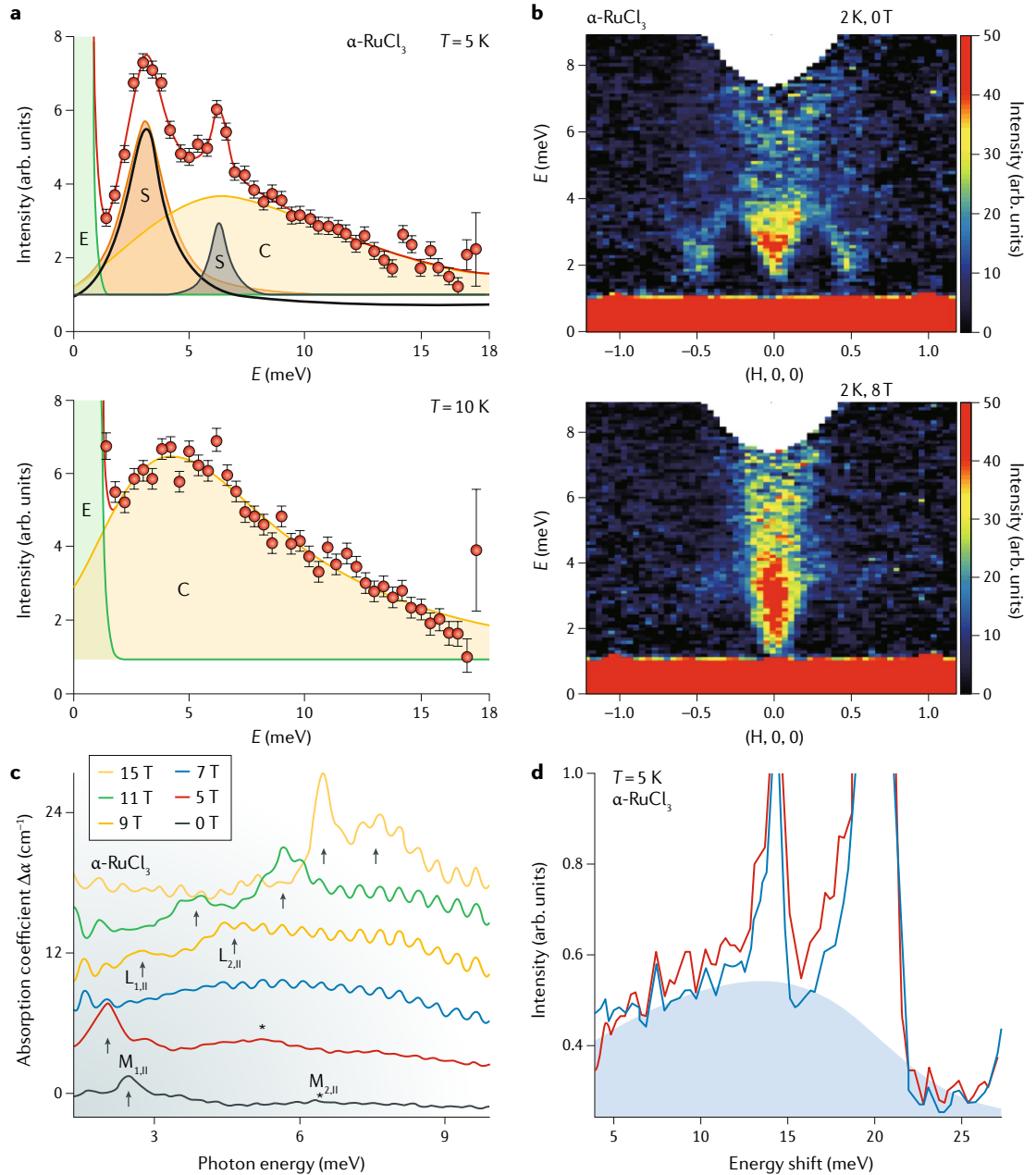


Fig. 6 | Signatures of fractional excitations and non-Kitaev interactions. a | Inelastic neutron scattering in single-crystal α -RuCl₃ measured at temperatures of $T = 5$ K, below the ordering temperature (shown in the upper plot), and 10 K, above the ordering temperature (shown in the lower plot). Data are integrated over a small reciprocal-space volume centred at the Γ point of the 2D lattice. The letters designate the contributions from the elastic line (E), spin waves (S), and continuum scattering (C). **b** | Inelastic neutron scattering is measured at $T = 2$ K along the reciprocal $[H, 0, 0]$ direction in a zero external magnetic field (shown in the upper image) and in a field of 8 T in the honeycomb plane (shown in the lower image), large enough to suppress the magnetic order. **c** | THz spectroscopy measurements in α -RuCl₃ subject to a magnetic field applied in the honeycomb plane, with the THz field parallel to the applied field direction. All measurements were carried out at $T = 2.4$ K. The arrows indicate locations of excitations inferred from the data. **d** | Detail of Raman measurements in α -RuCl₃ at $T = 5$ K. The blue curve is data from collinear (XX) polarization; the red line is data from cross (XY) polarization. The shaded area represents the magnetic continuum scattering. E, energy; L and M, modes. Panel **a** is reproduced with permission from REF.⁷⁶, AAAS. Panel **b** is reproduced from REF.⁹², CC-BY-4.0. Panel **c** is reproduced with permission from REF.⁹¹, APS. Panel **d** is reproduced with permission from REF.¹⁰⁸, APS.

response from the ferromagnetic and antiferromagnetic QSLs, there is even ambiguity over the sign of the Kitaev term needed within a Kitaev description⁹². Despite these uncertainties, the preponderance of experimental evidence suggests that the continuum scattering

seen in INS is a signature of fractional magnetic excitations, and these may be related to the excitations of a Kitaev QSL.

THz spectroscopy provides a high-resolution measurement of the response at zero momentum ($Q = 0$).

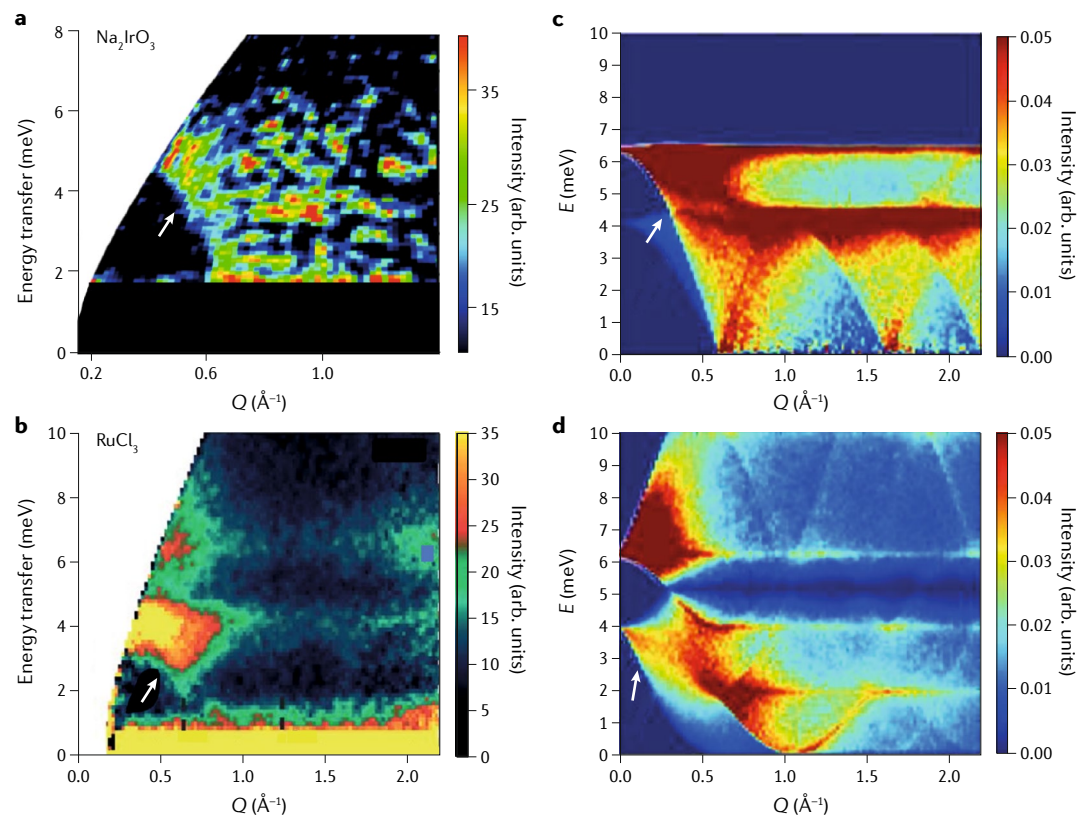
Box 2 | Inelastic neutron scattering measurements in polycrystalline materials

Single crystals of collinearly ordered magnetic systems have an inelastic neutron scattering (INS) cross section consisting of a sharp peak at the position of allowed spin-wave modes. The dispersion (that is, the wavevector dependence of the energies) and strength of the peaks can be analysed to infer the parameters relevant to a proposed model Hamiltonian.

When measurements are carried out on powders, the orientational information is lost, and the scattering is proportional to a weighted density of states. This yields less information but can still be useful. Neutron scattering intensity measured in a powder of Na_2IrO_3 (REF.⁵⁵) at $T = 4.6$ K, with scattering (Q) at energies below 2 meV suppressed for clarity, is shown in the figure, panel a. As discussed in the main text, it is experimentally challenging to obtain high-quality INS data on the iridate materials. Compounds based on Ru do not suffer from this complication, so measurements with much better statistics are possible, as illustrated by INS data from a powder of $\alpha\text{-RuCl}_3$ (REF.⁶⁴) (see the figure, panel b) at $T = 5$ K. Note that there is no normalization in the data sets plotted in the figure, panels a and b.

The powder scattering in both Na_2IrO_3 (in the figure, panel a) and $\alpha\text{-RuCl}_3$ (in the figure, panel b) shows a low energy (E) feature with a threshold shape that is concave towards the origin. This was interpreted as a signature of underlying zigzag order because a convex threshold shape is expected for spin waves from the stripy ground state for most plausible Hamiltonians describing the system^{55,64}. Representative calculations of the powder averaged scattering model (supplementary materials of REF.⁶⁴) is shown for both the zigzag (in the figure, panel c) and stripy (in the figure, panel d) models. The threshold is indicated by the white arrow in each case. The measured convex shape of the scattering threshold indicates zigzag order in both Na_2IrO_3 and $\alpha\text{-RuCl}_3$. The low-momentum scattering threshold is much more crisply defined in $\alpha\text{-RuCl}_3$.

The complete scattering in $\alpha\text{-RuCl}_3$ contains an additional feature at higher energies⁶⁴ that appears near 6 meV (in the figure, panel b). When the temperature is increased to $T > T_{\text{mag}}$ (T_{mag} is the ordering temperature, ~ 14 K in polycrystalline material), the lower-energy spin-wave scattering loses definition and softens dramatically. One would expect the scattering from a higher-energy spin-wave mode to be greatly diminished at temperatures larger than $2T_{\text{mag}}$. This is not the case for the feature seen in $\alpha\text{-RuCl}_3$, which has intensity persisting to temperatures of 100 K or more. This scattering arises from the powder average of the continuum scattering seen near the Γ point (see main text and the supplementary material of REF.⁷⁶ for a more detailed discussion). The overall energy width and T -dependence of this feature was seen to be consistent with scattering expected from fractional excitations⁶⁴. Panel a adapted with permission from REF.⁵⁵, APS. Panels b–d reproduced from REF.⁶⁴, Springer Nature Limited.



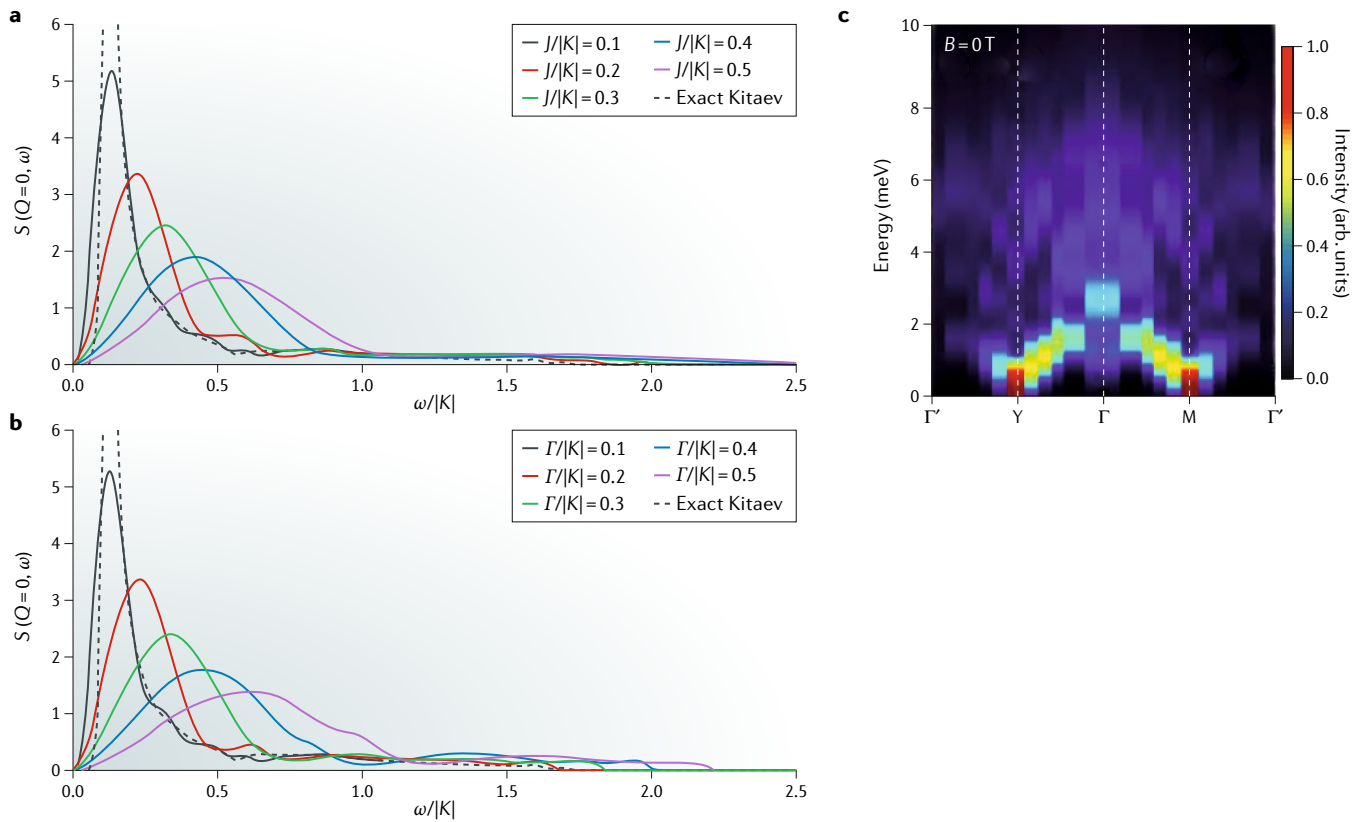


Fig. 7 | Response functions in extended Kitaev models. a,b | Structure factor $S(Q=0, \omega)$ at the Γ point calculated using perturbation theory from the pure Kitaev limit¹²¹. The Kitaev interaction, assumed equal along all three directions, is denoted by K . Panel **a** shows the effect of a perturbing Heisenberg term of strength J , and panel **b** shows the effect of a perturbing off-diagonal exchange term Γ . The most obvious effects of either perturbation are a broadening and renormalization of the low energy peak in the spectrum that, in the pure Kitaev model, is a signature of the static fluxes. **c** | Inelastic neutron scattering intensity calculated by exact diagonalization¹¹⁷ for a model Hamiltonian including Kitaev, off-diagonal and first-neighbour and third-neighbour Heisenberg terms. The parameter values used are representative of estimates proposed to describe α -RuCl₃ (REF. 123). Panels **a** and **b** are adapted with permission from REF. 121, APS. Panel **c** is adapted with permission from REF. 117, APS.

This complements INS measurements, because owing to kinematic constraints, INS can access the Γ point of the quasi-2D Brillouin zone only for non-zero values of the out-of-plane momentum. THz spectroscopy has been used to obtain detailed measurements of the spin waves at $Q=0$ (REFS^{113–115}) that provide crucial information about the full Hamiltonian describing α -RuCl₃. However, it appears that the weak inter-planar interactions are antiferromagnetic; thus, in the ordered state, the lowest gapped excitations at $Q=0$ may not represent the minimum spin gap associated with the 2D Γ point. Broadly speaking, the results from THz spectroscopy are consistent with those from INS. THz spectroscopy measurements at $T=2.4$ K over a range of magnetic fields extending up to 15 T (REF. 91) show spectra with sharp spin-wave peaks coexisting with a broad continuum at low fields, the vanishing of the spin waves at B_c and the emergence of a gapped mode at high fields, with energy increasing linearly with B (FIG. 6c). A spin gap linear in B is consistent with theoretical expectations for a system described by a Hamiltonian that is perturbed away from the Kitaev limit^{103,117} (BOX 2). Such a mode has also been observed in high-field ESR experiments¹¹⁰, albeit with a different energy to that seen with THz spectroscopy.

NMR measurements of the field dependence of the spin gap in α -RuCl₃ have been somewhat controversial. NMR is sensitive to the field-induced suppression of magnetic order⁹⁰, but at least one group of researchers has reported gapless excitations in the disordered state¹¹¹, whereas another finds the gap growing as B^3 at high fields¹¹². The latter result has been interpreted as arising from two-gauge-flux excitations such as those produced by a spin flip in the pure Kitaev model, with the temperature dependence of the local susceptibility and spin relaxation providing evidence for the expected additional Majorana fermion.

For the pure Kitaev model, the response function measured in INS contains contributions from pairs of static gauge-fluxes as well as one or more odd numbers of mobile Majorana fermions²⁹. By contrast, Raman scattering arises from a process involving pairs of Majoranas, enabling an easier calculation of the intensity^{30,124}. Low-temperature Raman scattering in α -RuCl₃ (REF. 108) (FIG. 6d) exhibits a continuum that is evidently magnetic and which cannot be explained by conventional two-magnon scattering but has a strong resemblance to the expected scattering from a pure Kitaev model. Further analysis of the temperature dependence

of the Raman signal provides additional evidence for the fractional nature of the underlying spin excitations³¹.

Half-quantized thermal Hall effect in α -RuCl₃.

Non-spectroscopic measurements of, for instance, thermal transport are another promising way to detect the fractionalization of spin excitations. The chiral edge state of the topological Kitaev QSL under a magnetic field is predicted to give rise to a half-quantized thermal Hall conductivity κ_{xy}/T . The half-integer factor arises from fractionalization into Majorana fermions, because Majorana fermions carry only half of the degrees of freedom of canonical fermions. A thermal Hall conductivity close to a half-quantized value $\kappa_{xy}/T = 1/2(\pi/6)(k_B^2/\hbar)$ was recently reported for single crystals of α -RuCl₃ with a magnetic field close to the critical in-plane value, B_c (REF.⁸⁹). Prior to this, an unusually large thermal Hall effect with applied magnetic field $B_\perp$ perpendicular to the plane (in-plane field $B = 0$) was discovered in α -RuCl₃ above T_N (REF.¹²⁵). It was discussed as a possible signature of a half-quantized thermal Hall effect, masked by the long-range ordering below T_N . The long-range order can be suppressed by superposing an in-plane field $B_\parallel$, as discussed above; this can be done by tilting the magnetic field from the perpendicular direction. The tilted-field data showed a plateau-like behaviour of κ_{xy}/T as a function of the applied field (FIG. 5b)⁸⁹. The κ_{xy}/T value on the plateau was independent of the tilting angle and close to the expectation for the half-quantized thermal Hall conductivity $\kappa_{xy}/T = 1/2(\pi/6)(k_B^2/\hbar)$. The in-plane component of magnetic field $B_\parallel$ in the plateau region was slightly larger than the critical field B_c , suggesting that the phenomena is closely linked to the B -induced suppression of the long-range magnetic ordering. At the time of writing, this striking observation⁸⁹ is too new to have been reproduced by other groups, and the full details have not been explored. The ultimate significance of the apparent fractional excitations in α -RuCl₃ and the relationship between the field-induced disordered state and the Kitaev QSL are at the forefront of current research.

Perspectives

The emergence of the exactly solvable Kitaev model a decade ago was a significant breakthrough in quantum magnetism. The impact of the Kitaev model was not confined to the theoretical community but soon found expression in correlated oxide physics, in which the concept of the spin–orbital Mott insulator had just been established, and to topological physics, in which the hunt for Majorana fermions became an important topic. This linkage grew into an interdisciplinary field whose main goal is the materialization of Kitaev QSLs.

As seen in this Review, the progress in the past few years is quite impressive. A number of honeycomb spin–orbital Mott insulators have been proposed to accommodate the essential ingredients of the Kitaev model. At the same time, a number of firm theoretical predictions for distinct physical properties of the Kitaev QSL, in particular those signifying the presence of Majorana fermions, have been proposed for the pure Kitaev model. Furthermore, a QSL ground state was established in H₃LiIr₂O₆ (albeit not yet identified as a Kitaev QSL), and the footprint of Majorana fermions may have been captured in α -RuCl₃ under magnetic fields.

Nevertheless, there is a gap between the material realizations and the theoretical models. The materials do not realize the pure Kitaev model but accommodate many other ingredients, such as additional exchange interactions, and these mask the manifestation of pure Kitaev physics. There is a need for realistic theories with additional ingredients to describe the QSL states and the elementary excitations. Open questions include what kind of QSL they are; whether there is any connection between the pure Kitaev model and other QSLs; and whether the elementary excitations in real materials can be described as Majoranas. There is also a need for more materials with the right ingredients for Kitaev physics, including possible candidates without Ir⁴⁺ and Ru³⁺. The game has just started.

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Author contributions

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Competing interests

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